HEC-RAS 2D Dynamic River Simulation Guide

Technical Documentation

May 18, 2026

1 Introduction

This document outlines the workflow for conducting a 2D unsteady flow (dynamic) simulation in HEC-RAS using a topobathymetric GeoTIFF and known water surface elevations (heads).

2 Part 1: Setting Up the Terrain

Before performing hydraulic calculations, HEC-RAS requires a Digital Elevation Model (DEM) in a specific format (.hdf).

2.1 Open HEC-RAS:

Create a new project.

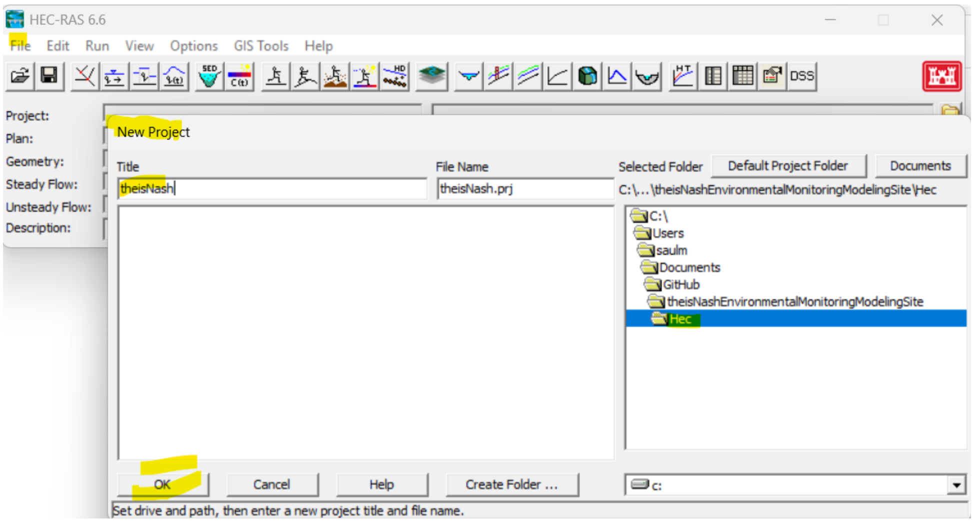

Figure 1: Create project

But be careful of the unit system

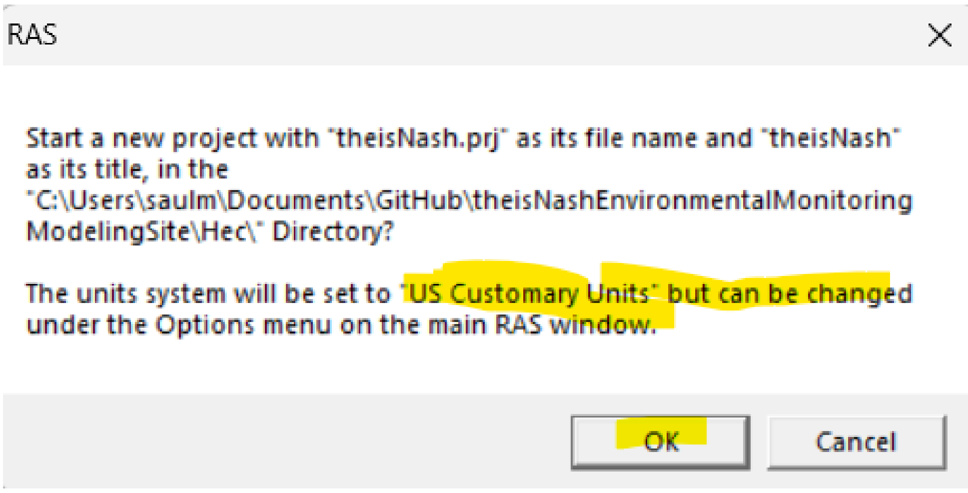

Figure 2: Warning over the unit system

Before importing shapefile its time to change the unit system of the project. Go to Options > Unit System (US Customary or System International).Select System International (SI): Meters, kilometers, etc.

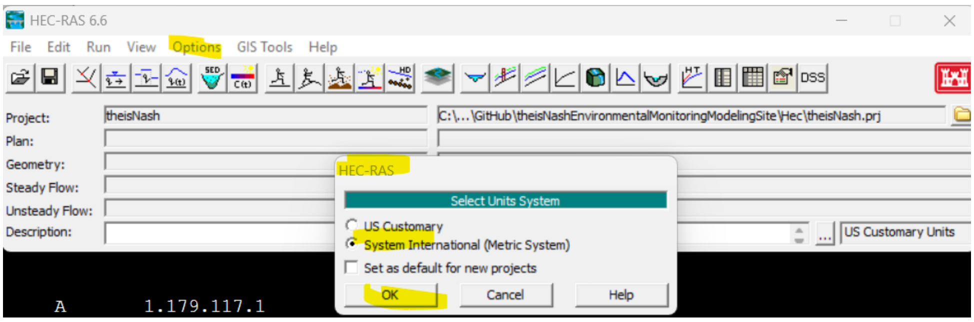

Figure 3: Select Units System

2.2 RAS Mapper:

Open the RAS Mapper tool.

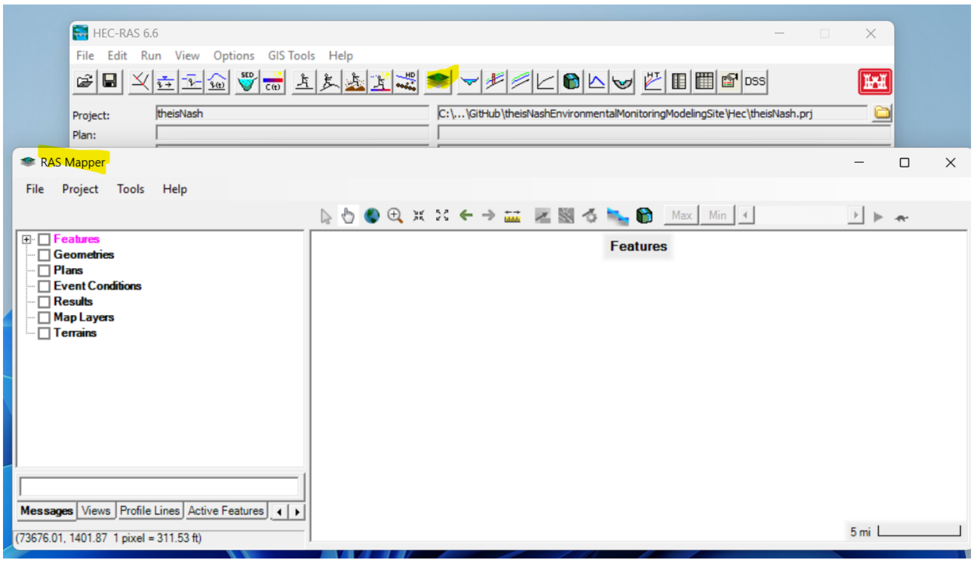

Figure 4: Ras Mapper tool

2.2.1 Coordinate System:

Go to *Project* > *Set Projection*. Select a Projection file (.prj) that matches your GeoTIFF's coordinate system.

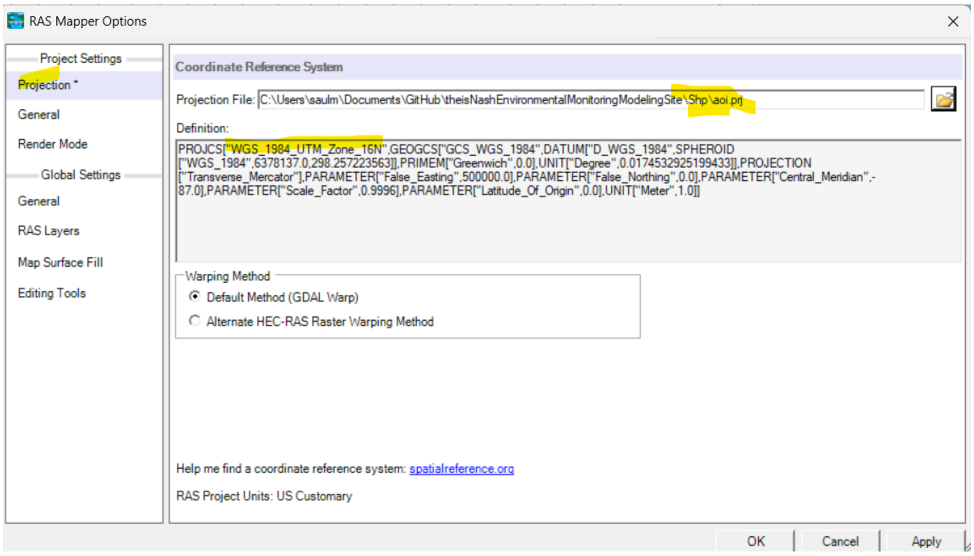

Figure 5: Set projection

2.2.2 Create Terrain:

Right-click on *Terrains* and select *Create New RAS Terrain*.

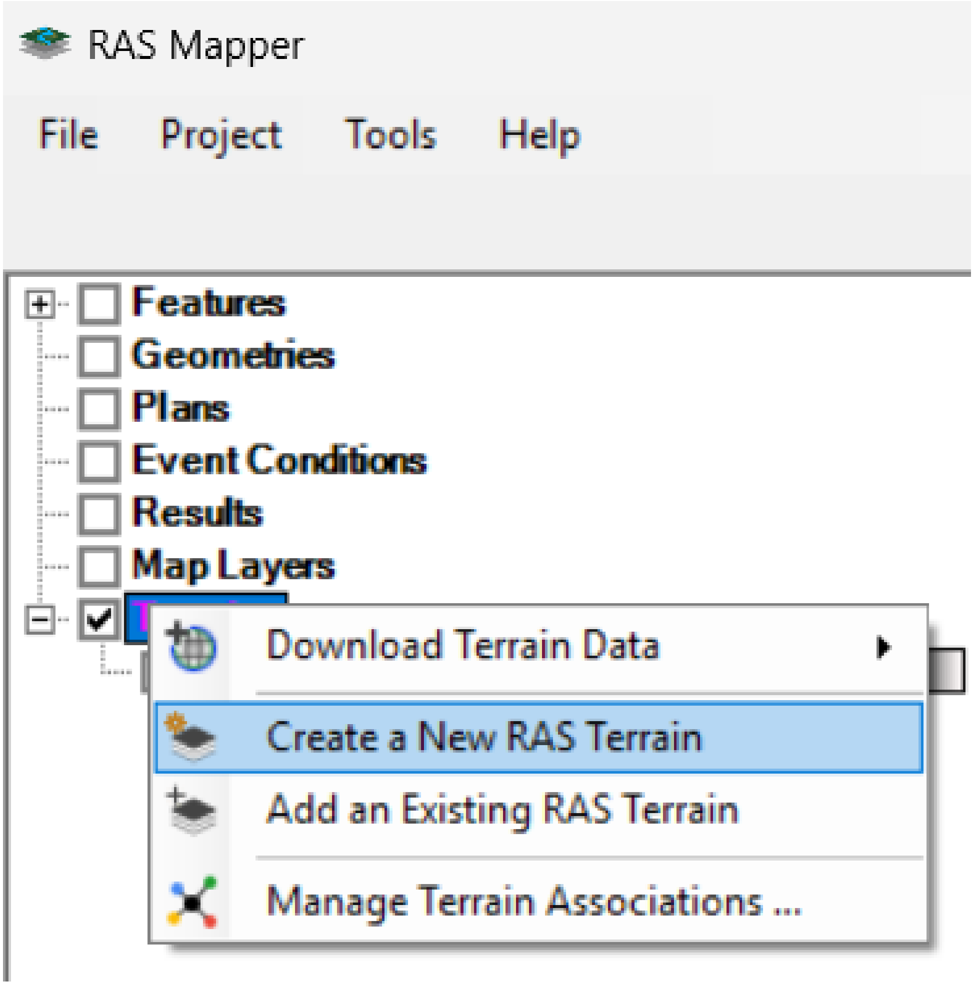

Figure 6: Create New RAS Terrain

Add your topobathymetric `.tif` file. This will convert the GeoTIFF into a HEC-RAS Terrain layer.

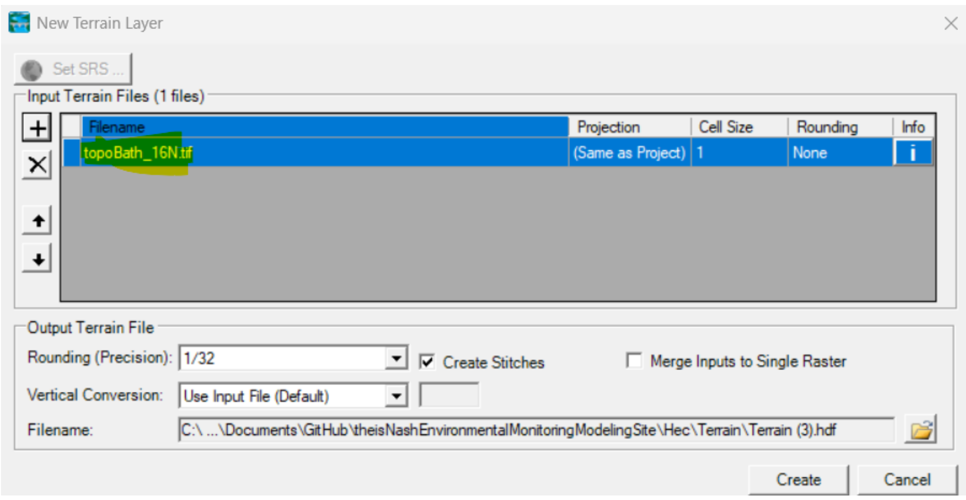

Figure 7: Add your tif file

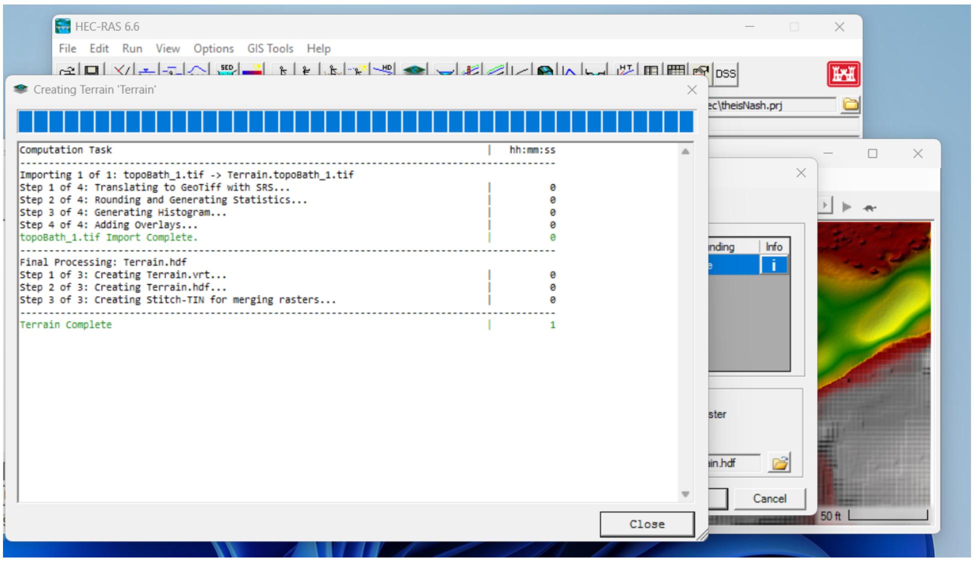

Figure 8: Hdf creation

3 Part 2: Defining the 2D Flow Area

3.0.1 Create Geometry:

In RAS Mapper, right-click *Geometries* and select *Add New Geometry*.

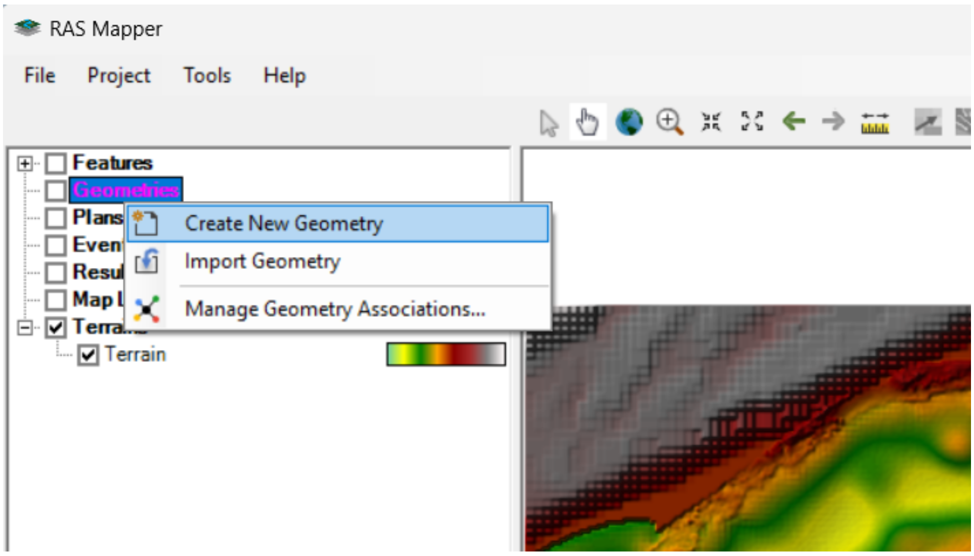

Figure 9: Add New Geometry

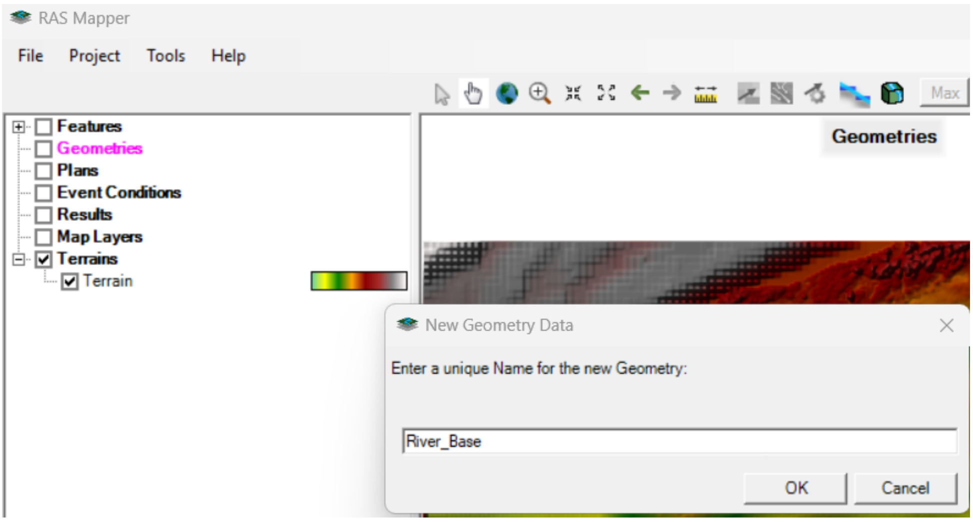

Figure 10: New Geometry Data

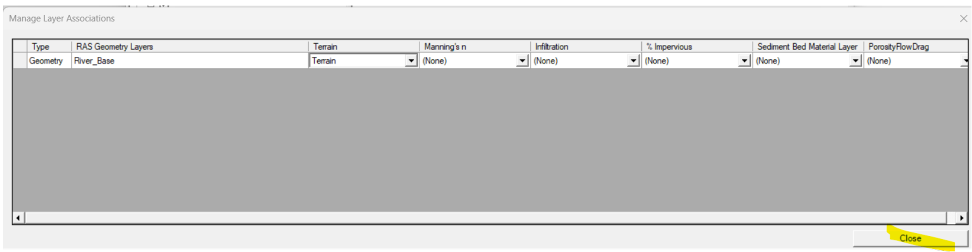

Figure 11: Manage Layer Associations

3.0.2 Import the Shapefile into a Layer

You cannot simply "open" a shapefile as a geometry; you must import its features into the specific HEC-RAS layer (like the 2D Flow Area / Perimeter).

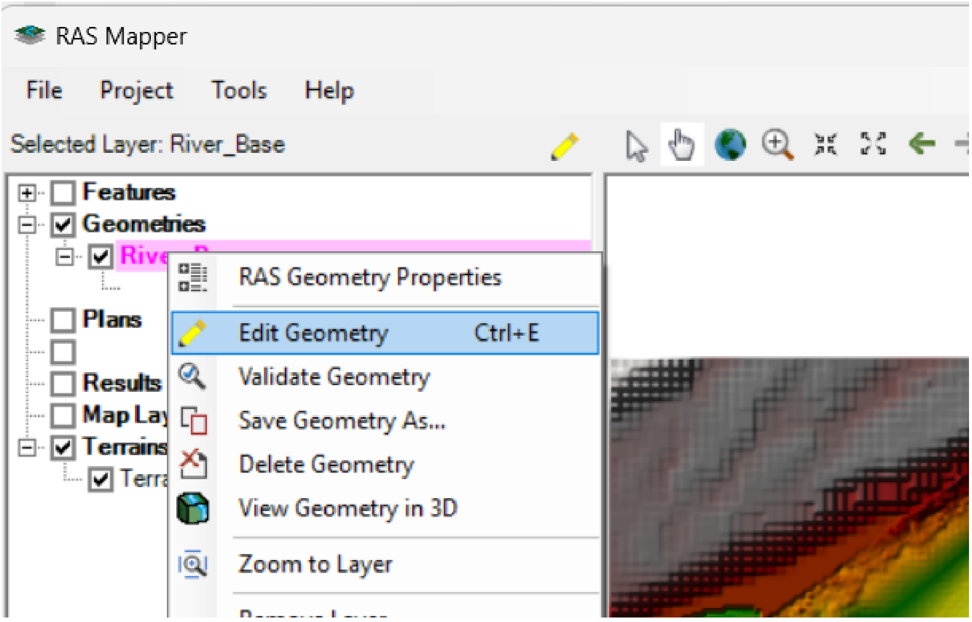

Figure 12: Edit geometry

3.0.3 Create Geometry:

Right-click on the specific sub-layer you want to populate (e.g., 2D Flow Areas / Perimeters).

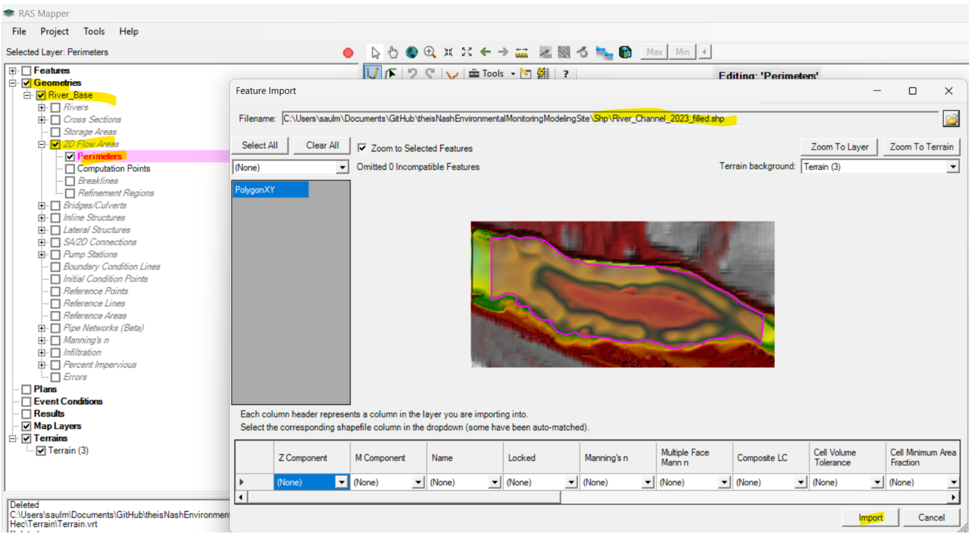

Figure 13: Feature Import

Right-click the 2D area polygon and select *Edit 2D Flow Area Properties*.

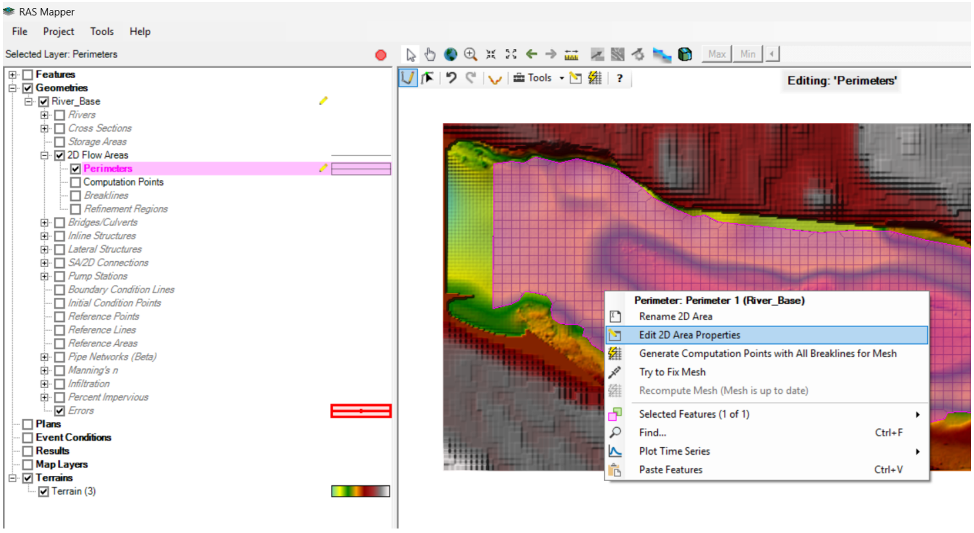

Figure 14: Edit 2D Flow Area Properties

Define your cell size (e.g., 5m x 5m). Click *Generate Computation Points*.

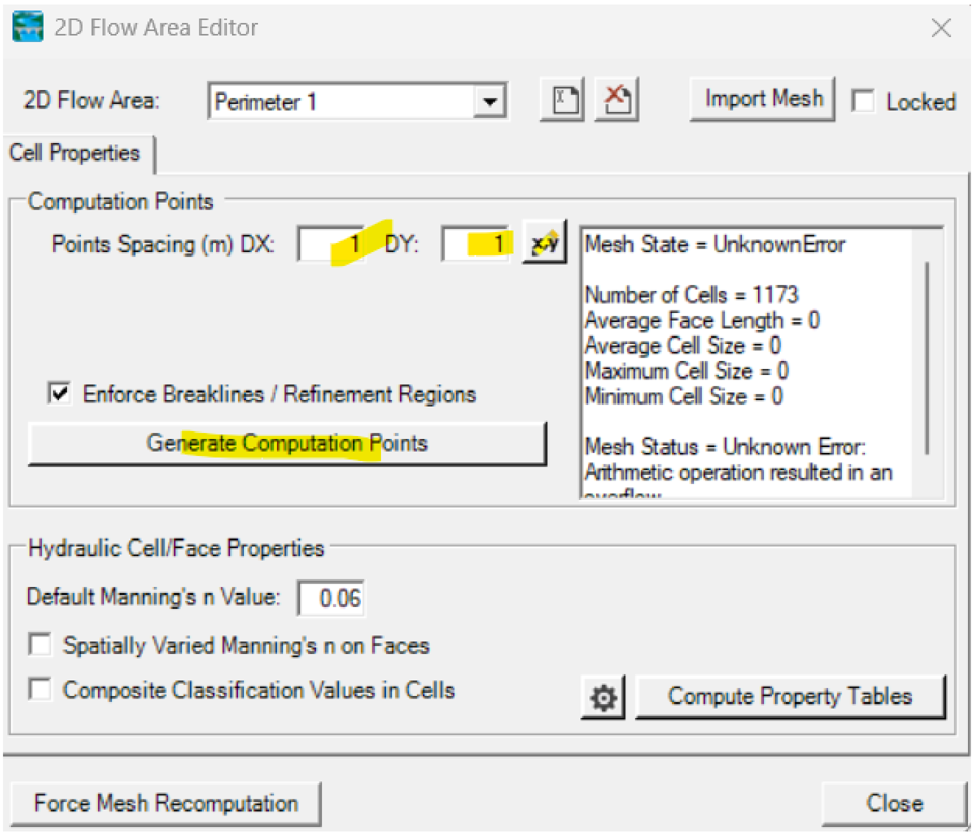

Figure 15: Edit 2D Flow Area Properties

4 Part 3: Boundary Conditions

Since you have "heads" (Water Surface Elevations - WSE), you will use these as boundary conditions.

4.1 Boundary Condition Lines

In RAS Mapper, select the *Boundary Condition Lines* layer under your Geometry.

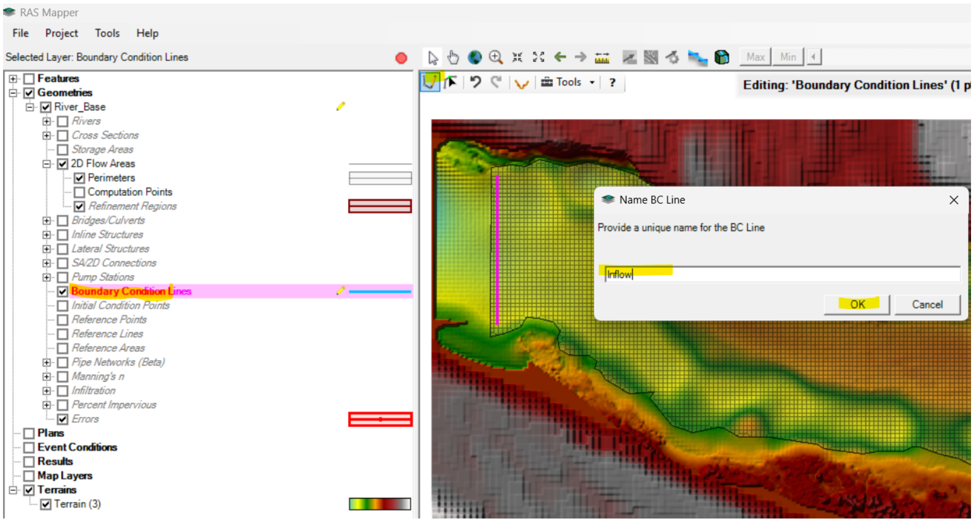

Figure 16: Name BC Line - Inflow

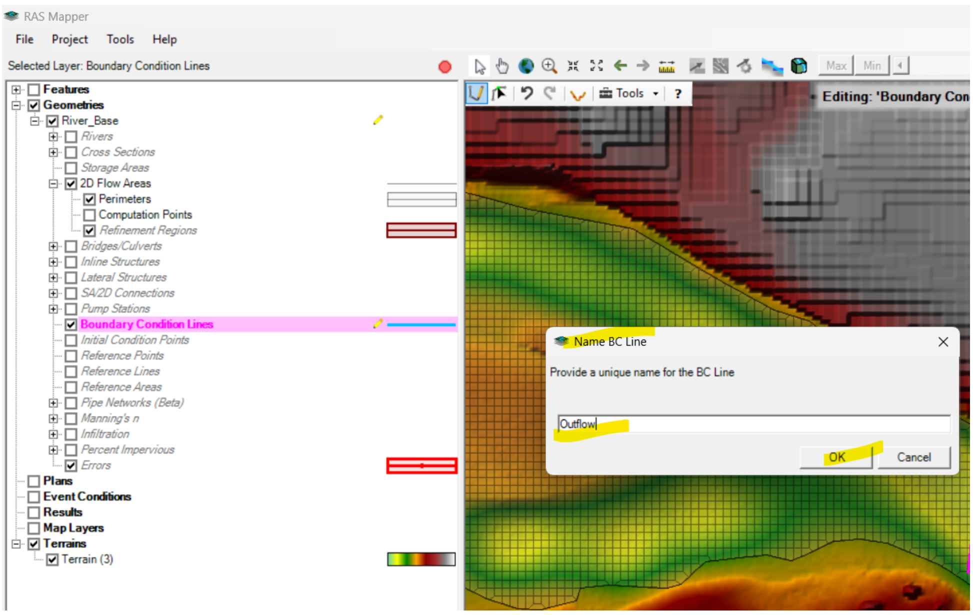

Figure 17: Name BC Line - Outflow

5 Part 4: Simulation of unsteady flow

You mentioned having "heads" (WSE) but want to know where the "flow" is. In a hydraulic model, flow (Q) and head (H) are related via the simulation.